

A. Foundational Concepts of Analytics

1. Business Analytics → Transforming raw data into actionable insights and analyse business performance.

Example: Can be used to see how well a product is doing in the store.

2. Data Analytics → The process of collecting, cleaning and organising data using Statistical tools and exploratory techniques.

Example: The data can be used to determine why there are negative reviews about the products.

3. Business Intelligence (BI) → When data is turned into dashboards and reports gathered from different sources like excel.

Example: Uses tools like powerBI to track new customers every month.

4. Data-Driven Decision Making → The use of data analysis to make informed decisions instead of relying on ones experience in the job or industry.

Example: Use data and insights to conclude why the performance has gone down in a business and not rely on work experience.

5. Big Data → Complex data that can only be stored and analyzed using advanced tools.

Example: Financial services like Banks.

6. Descriptive Analytics → Summarises ^{past} data to understand what happened.

Example: A store analyzing historical data and create a report to understand their sales in the past 2 months.

7. Diagnostic Analytics → Explains why something happened by identifying patterns and relationships in the data.

Example: An online retail store website gets 100 visits and post insights gathering they find out that this ^{is} caused by ~~the~~ slow website response on mobile.

8. Predictive Analytics → Uses data and models to forecast future outcomes

Example: using historical data to find out why customers are leaving the company by identifying monthly patterns

9. Prescriptive Analytics → Recommends actions based on insights from predictive analytics

Example: A wellness spa analyzes customer booking behaviour to enhance their packages

10. Data Literacy → The ability to read, understand and work with data to make informed decisions

Example: A dashboard that monitors website traffic.

B. Data, Measurement & Reporting

11. Structured data → Data stored in a table with rows & columns

Example: - Data stored in a spreadsheet.

12. Unstructured data → ~~less~~ Inconsistent data.

Example: - A social media post.

13. Data warehouse → Central place used to store databases

Example: - In banking, it's used to collect data from several sources (ie customer details, customer transactions)

14. Data mining → process of analysing patterns and relationships using machine learning in large datasets.

Example: - can be used in banking to see which customers have less chances of loan approval because of bad credit score ~~data~~ ^{or} missed payments.

15. Data ~~data~~ visualization → Representation of data through dashboard charts

Example: A pie chart to showcase an expense breakdown

16. Dashboard → visual interface that shows key performance indicators to support decision making

Example: - policies acquired monthly.

17. Metric $\rightarrow$ A number that tracks performance.

Example:- ~~As~~ Visits on the Amazon website.

18. Key Performance Indicator (KPI) $\rightarrow$ A metric that measures success against set goals.

Example:- Return on Investment (ROI)

19. Benchmark $\rightarrow$ Valuable business analysis technique to improve processes.

Example:- Use benchmarking to assess your organization's performance against industry competitors.

20. Segmentation $\rightarrow$ Dividing large group of data into smaller groups.

Example:- A store groups its employees into small groups based on their buying behavior. i.e. returning buyers and one-time buyers.

C. Customer & Sales KPIs

2.1 Conversion Rate $\rightarrow$ percentage of users that complete a desired action

$$CR = \frac{\text{Number of conversions}}{\text{Total visitors}} \times 100$$

Example:- 3000 website visitors and 400 users purchased.
 $(400 \div 3000) \times 100 = 13.4\%$

2.2 Customer Acquisition Cost $\rightarrow$ total cost a business spends to gain a new customer. $CAC = \frac{\text{Total sales and marketing costs}}{\text{Number of new customers acquired}}$

Example:- R50 000 total costs and 300 new customers.

$$CAC = \frac{R50\,000}{300} = R167 \text{ per customer.}$$

2.3 Customer Lifetime Value (CLV/LTV) $\rightarrow$ measures the total ^{revenue} ~~as~~ a business ~~conceives~~ ^{expects} from a single customer throughout their entire relationship.

Example:- $CLV = \text{Average Purchase Value} \times \text{Purchase Frequency} \times \text{Customer Lifespan}$
 $CLV = 400 \times 2 \times 3 = 2400$

24. Churn Rate $\rightarrow$ Percentage of customers who stop using a company's product over a specific period.

Example:- 50,000 Spotify subscribers and 400 subscribers cancelled membership
 $CR = (\text{number of customers lost} \div \text{Total customers at the start})$

25. Customer Retention Rate $\rightarrow$ How good a business keeps its customers.

Example:- 3,000 customers beginning of the month. During the month 200 new customers join and 100 remain at month.

$$CRR = \frac{\text{Customers at end} - \text{new customers}}{\text{Customers at start}} \times 100$$

$$CRR = \frac{(1000 - 200)}{3000} \times 100$$

$$CRR = \frac{800}{3000} \times 100 = 27\%$$

26. Net Promoter Score (NPS) $\rightarrow$ Metric used to measure customer satisfaction.

Example:- How likely are you to recommend Bata app. Answer on a scale of 0-10.

27. Average Order Value - Amount spent by customers everytime they make a purchase.

Example:- $AOV = \frac{\text{Total revenue}}{\text{Number of orders}}$

28. Sales funnel $\rightarrow$ A model that shows how customers gain interest in a product to buying it.

Example:- Pop up ads on social media.

D. Marketing & Campaign Analytics

29. Marketing Campaign $\rightarrow$ A strategy to convert potential customers into buying customers.

Example:- A Pop back to school uniform advert.

30. Post-campaign Analysis → process of reviewing a marketing campaign after it has ended.

Example:- Review the number of people it reached and the engagement.

31. Incremental Revenue → Additional revenue generated by a specific campaign.

Example:- A retail store sells 3 t-shirts for R60 ^{R10} and 10 dresses at R150.

32. Return on Investment → financial metric used to measure how much money was gained / lost in an investment.

Example:- $ROI = \frac{\text{Profit} - \text{Investment cost}}{\text{Investment cost}} \times 100$

33. Return on Ad spend → Measures every ^{rand} spent on advertising.

Example:- $ROAS = \frac{\text{Revenue from ads}}{\text{Ad Spend}}$.

34. Click-Through Rate → How effective an advertisement was through viewer engagement.

Example:- $CTR = \frac{\text{Total measured Ad clicks}}{\text{Total Measured Ad impressions}} \times 100$

35. Bounce Rate → Users in percentage that visited a website with no action.

Example:- $BR = \frac{\text{Single-page visits}}{\text{Total visits}} \times 100$

36. A/B Testing → Comparing two forms of an advertisement to see which one performs better.

Example:- Create two forms of adverts and divide your audience and see which one performs better and use a Click through rate to measure ^{Success} ~~performance~~.

37. Lead Generation → The process of categorising potential customers for your business and guiding them into your sales.

Example:- Get users to buy products on your online retail store.

38 Marketing Attribution -> Identifying which efforts contributed to sales increase.

Example:- A customer purchases an item on the Bash app after seeing an advert pop up on Facebook.

References

1. Tim Mucci. IBM writer. (2026). What is data-driven decision-making? <https://www.ibm.com/think/topics/data-driven-decision-making>.
2. By Adam Hayes. Reviewed By Margaret James. Investopedia. (2026). ~~What is a Marketing Strategy?~~ Understanding Click-Through Rate (CTR): Definition, Formula, and Importance. <https://www.investopedia.com/terms/c/clickthroughrates.asp>
3. David Kindness. The Investopedia Team. (2026). What is a Marketing Strategy
4. Alexandra Jorner; Alice Elmstyn. IBM Think. (2026). What is data literacy? <https://www.ibm.com/think/topics/data-literacy>.
5. Hermant Manral. International Institute of Business Analysis. Business Analysis Technique: Benchmarking.